EASY CLAIM – AI ASSISTED INSURANCE CLAIM FILLING PLATFORM

Submitted in partial fulfilment of the

requirements of the degree of

Bachelor of Engineering (B.E.)

in

INFORMATION TECHNOLOGY

by

Arya Uday Kurup	**EU2224008**
Abhijeet Pradip Rogye	**EU2224002**
Viraj Atul Tamhanekar	**EU2224004**

Under the guidance of

Mr. Gautam Jha

Assistant Professor

Department of Information Technology

St. John College of Engineering and Management, Palghar

University of Mumbai

2025–2026

CERTIFICATE

This is to certify that the B.E. project entitled **"Easy Claim – An AI Assisted Insurance Claim Filling Platform"** is a bonafide work of **Arya Uday Kurup (EU2224008)**, **Abhijeet Pradip Rogye (EU2224002)**, **Viraj Atul Tamhanekar (EU2224004)** submitted to University of Mumbai in partial fulfilment of the requirement for the award of the degree of **"Bachelor of Engineering"** in **"Information Technology"** during the academic year 2025–2026.

Mr. Gautam Jha
Project Guide

Dr. Arun Saxena	**Dr. Kamal Shah**
Head of Department	Principal

B.E. Project Report Approval

This project report entitled **"Easy Claim – An AI Assisted Insurance Claim Filling Platform"** by ***Arya Uday Kurup, Abhijeet Pradip Rogye,*** and ***Viraj Atul Tamhanekar*** is approved for the degree of ***Bachelor of Engineering*** in ***Information Technology*** from ***University of Mumbai***.

Examiners

1.________________________________

2.________________________________

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

1. Arya Uday Kurup (EU2224008)

   Signature: ____________________

2. Abhijeet Pradip Rogye (EU2224002)

   Signature: ____________________

3. Viraj Atul Tamhanekar (EU2224002)

   Signature: ____________________

Date: ________________

Place: _______________

ABSTRACT

The traditional insurance claim filing process remains a fragmented and time-consuming administrative challenge, often characterized by extensive manual paperwork and a complete lack of structured user guidance. This project presents **Easy Claim**, an AI-assisted insurance claim management system engineered to automate the end-to-end claim submission and tracking cycle.

By integrating **Optical Character Recognition (OCR)**—specifically via AWS Textract—and **Natural Language Processing (NLP)**, the platform intelligently extracts critical data from unstructured medical bills and prescriptions. This data is then mapped directly to predefined insurance form fields, effectively eliminating the risk of human error during manual entry. Furthermore, the system utilizes **Robotic Process Automation (RPA)** to simulate human interactions with external portals, such as **ECHS**, ensuring a seamless and rapid filing process.

System performance was evaluated through a structured dataset of insurance records, focusing on workflow optimization and the precision of data extraction. Experimental findings indicate a significant improvement in overall operational efficiency, with a recorded processing time reduction of approximately **84.8%** compared to conventional manual methods. The platform fundamentally enhances the user experience through interactive guided workflows, real-time status notifications, and a centralized management dashboard. By bridging the gap between legacy insurance procedures and modern digital automation, **Easy Claim** offers a scalable, user-centric, and highly effective solution for the contemporary digital insurance ecosystem.

Keywords: Insurance Claim Automation, Artificial Intelligence, OCR, NLP, RPA, Claim Tracking, Digital Insurance Platform, User Guidance.

Table of Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
OCR	Optical Character Recognition
NLP	Natural Language Processing
ML	Machine Learning
RPA	Robotic Process Automation
InsurTech	Insurance Technology
ECHS	Ex-Servicemen Contributory Health Scheme
CGHS	Central Government Health Scheme
AWS	Amazon Web Services
S3	Simple Storage Service
JWT	JSON Web Token
API	Application Programming Interface
REST	Representational State Transfer
DFD	Data Flow Diagram
UAT	User Acceptance Testing
UML	Unified Modeling Language
IDE	Integrated Development Environment
ESM	Ex-Servicemen
CSS	Cascading Style Sheets
HTTPS	HyperText Transfer Protocol Secure

CHAPTER 1

INTRODUCTION

The insurance industry, while fundamental to healthcare and economic stability, has historically struggled to keep pace with the rapid digital transformation seen in other financial sectors. Even in 2026, many policyholders find themselves navigating a frustrating maze of manual paperwork, delayed approvals, inconsistent communication, and fragmented digital portals that rarely deliver true convenience. What should be a straightforward reimbursement process often turns into a time-consuming ordeal, especially during moments when users are already dealing with medical stress.

A major challenge lies in the lack of integration across systems. Hospitals, insurers, and third-party administrators often operate in silos, forcing users to repeatedly submit the same documents, track claim statuses across multiple platforms, and deal with unclear or delayed responses. This not only reduces efficiency but also erodes trust in the system. Despite the availability of digital tools, most existing solutions fail to provide end-to-end automation or meaningful transparency. Easy Claim is designed to dismantle these long-standing barriers by introducing a unified, intelligent platform that simplifies and automates the entire claim lifecycle. From the moment a medical bill is received to the final settlement notification, the system acts as a proactive assistant rather than a passive portal. It eliminates redundant steps, reduces dependency on manual intervention, and ensures users always know what's happening with their claim.

Beyond automation, the platform emphasizes transparency and user experience. Real-time status updates, intelligent notifications, and a centralized dashboard ensure that users are never left in the dark. Instead of chasing approvals, users can rely on the system to guide them through each step, flag missing information, and even predict potential delays before they happen.

Ultimately, Easy Claim transforms what has traditionally been a tedious administrative burden into a seamless, efficient, and user-centric experience. By bridging the gap between technology and healthcare insurance, it not only improves operational efficiency for providers but also restores confidence and convenience for policyholders—setting a new standard for how insurance claims should be managed in the digital age.

1.1 Motivation

The primary motivation behind Easy Claim stems from the visible frustration of users who, during stressful medical or personal emergencies, are forced to navigate complex claim procedures without adequate guidance. Traditional systems are often so cumbersome that policyholders occasionally drop legitimate claims simply due to confusion or the absence of real-time support. There is a profound opportunity to bridge this gap using Insurtech innovations that can automate repetitive tasks, such as extracting data from unstructured medical reports and prescriptions.

Seeing that the integration of RPA can reduce filing times by nearly **70%** and OCR can reach accuracy levels over **90%**, the motivation was to build a tool that doesn't just digitize the process but fundamentally re-engineers it for speed and reliability. Furthermore, this project is motivated by a commitment to the United Nations Sustainable Development Goals (SDGs), specifically promoting economic growth through innovation (SDG 8 & 9) and reducing inequalities by making insurance management accessible to those with minimal technical knowledge (SDG 10).

1.2 Problem Statement

The current landscape of insurance claim processing is marked by several critical inefficiencies that hinder both user experience and operational effectiveness. Claim settlement continues to rely heavily on manual data entry and physical document uploads, creating significant delays and administrative burdens for hospitals and insurers alike. At the same time, transparency gaps leave claimants with little to no visibility into the progress of their claims, which gradually erodes trust in the system.

These processes are further complicated by high error rates, as manual verification is prone to human mistakes that often result in incorrect data entry and unnecessary claim rejections. To make matters worse, most existing platforms offer only generic support, failing to provide intelligent, case-specific guidance tailored to an individual's policy details and submitted documents, ultimately leading to frustration and inefficiency across the entire claim journey.

1.3 Objectives

The core objective of Easy Claim is to develop a comprehensive, AI-assisted platform that simplifies the insurance experience through five strategic goals:

- To Automated Data Extraction: Utilize OCR (AWS Textract) and NLP to automatically extract structured information from medical bills and prescriptions, minimizing manual input.
- Seamless Portal Interaction: Use RPA scripts (via Playwright or UiPath) to simulate user actions, automatically filling out forms on legacy insurance portals like CGHS.

- Real-Time Transparency: Provide a centralized, interactive dashboard that allows users to track claim progress, view error logs, and receive instant notifications.
- Enhanced Accessibility: Design a user-friendly interface that offers step-by-step AI guidance, ensuring the platform is usable for individuals across various technical backgrounds and demographics.

1.4 Scope

The scope of this project encompasses the design, development, and evaluation of an end-to-end web and mobile-based platform. Technically, it includes a React.js frontend for an intuitive user experience and a Python (FastAPI/Flask) backend for robust data processing. The system is built to handle multiple insurance categories, including Vehicle, Health, Travel, and Property insurance. It specifically targets healthcare providers, insurance companies, and corporate sectors that manage high volumes of sensitive claim data. While the current prototype focuses on structured digital workflows and anomaly detection, the scope is designed to be scalable, allowing for future integration with Blockchain for immutable records and predictive analytics to forecast approval probabilities.

The platform is designed with modular architecture and API-first principles, enabling seamless integration with hospital management systems, insurer databases, and third-party verification services. It also aims to incorporate role-based access control, audit trails, and encryption mechanisms to meet regulatory requirements and maintain data integrity. From a user perspective, the system will evolve to include personalized dashboards, intelligent recommendations, and multilingual support to improve accessibility and adoption across diverse user groups. Furthermore, continuous evaluation through user feedback, performance metrics, and usability testing will drive iterative improvements, ensuring the platform remains efficient, reliable, and aligned with real-world operational needs.

1.5 Summary

In summary, Chapter 1 has established the foundational need for **Easy Claim** as a response to the antiquated and fragmented nature of current insurance processes. By identifying the core problems—ranging from manual delays to lack of transparency—this introduction has outlined a roadmap for a system that leverages **AI and Automation** to restore efficiency and trust. The objectives defined here focus not only on technical accuracy (aiming for a validation accuracy of **90.7%**) but also on the "human" element of the process through guided support and real-time alerts.

CHAPTER 2

REVIEW OF LITERATURE

2.1 “AI-Powered Document Processing: The Foundation of Automation” [1]

Research in the field of insurance technology has identified a significant paradigm shift toward the digitization of health records and administrative tasks using **Optical Character Recognition (OCR)** and **Natural Language Processing (NLP)**. A foundational study by Larumoce (2021) proposed an AI-based system designed specifically for extracting claim details from scanned documents, which directly addresses the inefficiencies of manual data entry. Similarly, Smith et al. (2021) conducted a comprehensive review of OCR and NLP applications, demonstrating how these technologies can transform unstructured medical reports into actionable data.

Despite Building on these findings, recent advancements in AI-driven workflow orchestration highlight the importance of integrating document intelligence with real-time decision-making systems. Emerging research emphasizes that combining OCR and NLP with automated validation and rule-based engines can significantly reduce claim processing time while improving accuracy and consistency.

2.2 “Machine Learning in Fraud Detection: Securing the Ecosystem” [2]

Fraudulent claim submissions remain one of the most critical challenges for insurance providers, leading to substantial financial losses and increased premiums for honest policyholders. Research by Chen et al. (2022) highlighted the effectiveness of **Supervised Machine Learning models** in identifying anomalies within claim data to prevent fraud. Further systematic reviews have evaluated how pattern recognition and data analysis can flag irregular claiming behaviors that would otherwise go unnoticed during manual reviews. Despite these advancements, many existing fraud detection frameworks are restricted to backend analysis and lack a transparent interface for users to understand how their claims are being verified. This highlights a need for platforms that combine intelligent anomaly detection with real-time user guidance to build trust within the system.

In response to these limitations, emerging approaches are exploring the integration of explainable AI (XAI) into fraud detection systems, enabling greater transparency in how decisions are made. By providing users with clear insights into why a claim may be flagged or delayed, such systems can reduce confusion and improve trust between policyholders and insurers.

2.3 “Robotic Process Automation (RPA): Bridging Legacy Gaps” [3]

As insurance providers transition to digital models, many still rely on "legacy" web portals that do not support modern API integrations. **Robotic Process Automation (RPA)** has emerged as a crucial bridge in this transition, replicating human interactions with web interfaces to automate repetitive data entry tasks. Gupta et al. (2023) examined the role of RPA in **InsurTech**, noting its ability to significantly reduce operational costs and human error. However, isolated RPA implementations often struggle with complex, non-standardized documentation. The **Easy Claim** platform addresses this limitation by using RPA in synergy with **AWS Textract** and **NLP-driven form mapping**, ensuring that automated portal interactions are based on accurately parsed data.

To further enhance this capability, recent developments suggest combining RPA with intelligent document processing (IDP) and adaptive learning models, allowing systems to handle diverse and unstructured document formats more effectively over time. By continuously learning from new data patterns and user interactions, such systems can improve accuracy in data extraction and mapping, even when faced with inconsistent formats. This evolution moves RPA beyond simple task automation toward a more cognitive automation framework.

2.4 “Blockchain for Secure and Immutable Insurance Data” [4]

In the current era of heightened data privacy concerns, the security of sensitive medical and financial information is paramount. Recent literature, such as the study by Tariq et al. (2025), explores the integration of Blockchain technology to create immutable records of insurance transactions, thereby enhancing transparency and accountability. By using Ethereum smart contracts or distributed ledger systems, insurance platforms can ensure that claim records are tamper-proof and auditable. While blockchain provides a robust security layer, research indicates it is often studied in isolation from AI-driven processing. A comprehensive solution must therefore look toward a hybrid model that secures data via blockchain while using AI to maintain the speed and efficiency required for real-time claim settlements.

Building on this perspective, emerging research emphasizes the importance of integrating blockchain with privacy-preserving techniques such as encryption, zero-knowledge proofs, and secure multi-party computation to ensure that sensitive data remains protected even within distributed systems. Such hybrid architectures not only safeguard data integrity but also enable controlled access.

2.5 "Intelligent Document Verification Pipelines" [5]

The authenticity of uploaded documents—such as surgery bills and doctor prescriptions—is a prerequisite for any automated claim settlement system. Li and Kumar (2020) developed an AI-based verification pipeline that validates the legitimacy of claim documents through structured data checks. Such systems are vital for reducing the administrative workload on claim assessors by filtering out incomplete or invalid submissions before they reach the review stage.

Building on this evolution, recent research highlights the emergence of multimodal and context-aware AI assistants that can process not just text, but also documents, images, and user history simultaneously. These systems are capable of guiding users through complex workflows—such as uploading documents, identifying missing information, and suggesting corrective actions—in real time.

2.6 "Document Processing: The Digitization of Health Records" [6]

Research in the field of insurance technology has identified a significant paradigm shift toward the digitization of health records and administrative tasks using Optical Character Recognition (OCR) and Natural Language Processing (NLP). A foundational study by Larumoce (2021) proposed an AI-based system designed specifically for extracting claim details from scanned documents, which directly addresses the historical inefficiencies of manual data entry. Similarly, Smith et al. (2021) conducted a comprehensive review of OCR and NLP applications, demonstrating how these technologies can transform unstructured medical reports into actionable data for insurance systems. The evolution of these technologies has made it possible to interpret a wide variety of medical documents, such as bills and prescriptions, converting them into structured formats that the Easy Claim platform utilizes to eliminate repetitive human labor.

2.7 "Cloud-Based Document Management and Scalable Architectures" [7]

The modern digital insurance ecosystem relies heavily on cloud infrastructure to manage the high volume of documents generated during the claim process. Research into scalable architectures emphasizes the use of object storage solutions, such as AWS S3, to provide durable and secure backups of medical documentation. Furthermore, the transition toward FastAPI and Flask for backend orchestration allows for the development of high-performance APIs that can handle complex OCR requests and storage tasks simultaneously. By utilizing cloud-based three-tier architectures, systems can ensure that claim data is not only processed quickly through AWS Textract but also stored in a manner that is accessible via responsive web and mobile interfaces.

CHAPTER 3

REQUIREMENTS ANALYSIS

3.1 Hardware Requirements

Sr. No.	Component	Specification
1	Processor	Intel Core i5 or above (Dual-core minimum; Quad-core for enterprise)
2	RAM	Minimum 8 GB (16 GB recommended for enterprise deployment)
3	Storage	256 GB SSD (SSD is prioritized for faster data retrieval)
4	Operating System	Windows 10/11, macOS, or Linux
5	Display	1280x720 resolution or higher
6	Internet	Stable broadband (Required for cloud-based AI services and S3 storage)
7	Input Devices	Keyboard, Mouse
8	Browser	Latest versions of Chrome, Firefox, or Edge

Table 1: Hardware Requirement Table

3.2 Software Requirements

Sr. No.	Component	Specification
1	Programming (Frontend)	JavaScript (ES6+), HTML, CSS
2	Programming (Backend)	Python
3	Frontend Framework	React.js
4	Backend Framework	FastAPI or Flask
5	Database	PostgreSQL or MongoDB
6	AI / OCR Engine	AWS Textract

7	Automation (RPA)	PlayWright or UiPath
8	Data Visualization	Pandas, NumPy (for claim insights)
9	Storage (Cloud)	AWS S3 (for document backups)
10	ML Libraries	Scikit-learn (for Decision Trees and Logistic Regression)
11	IDE	VS Code
12	Deployment	Docker or Cloud-based Virtual Machines

Table 2: Software Requirement Table

3.3 Functional Requirements

1. **User Registration and Authentication**

   The User Registration and Authentication system must provide secure login and account management. It should support unique identifiers for each user to link their claim history and policy details.

2. **Claim Initiation**

   Users must be able to initiate a new claim through a simple and intuitive interface. The system should offer predefined insurance categories such as Health, Vehicle, and Travel to streamline the selection process. It should guide users in choosing the correct category based on their claim type. This ensures faster and more accurate claim initiation.

3. **AI-Assisted Form Filling**

   The system must use OCR technology (AWS Textract) to extract relevant data from uploaded documents like medical bills and prescriptions. NLP techniques should further process this data to match required form fields. This reduces manual data entry and minimizes user errors. It also improves speed and overall user experience.

4. **Document Upload & Management**

   Users should be able to upload multiple supporting documents in various formats securely. Each document must be stored and linked to a unique Claim ID for easy retrieval. The system should ensure proper organization and accessibility of uploaded files. Secure storage mechanisms must be implemented to protect sensitive data.

5. **Automated Portal Submission (RPA)**

   The core execution logic of the platform relies on **Robotic Process Automation (RPA)** to interact directly with external insurance provider portals like **ECHS** or **CGHS**. The system must use automated scripts (via Selenium or UiPath) to log into these portals, navigate to the submission tabs, and populate the forms using the AI-extracted data. This ensures a rapid, consistent, and hands-free submission process

6. **AI Guidance Assistant**

   An AI-driven assistant should guide users step-by-step throughout the claim filing process. It must provide personalized suggestions and real-time feedback based on user inputs. The assistant can help reduce errors and improve form completion rates. This ensures a smooth and user-friendly experience.

7. **Historical Claim Logs and Audit Trail**

   The system must maintain a comprehensive and immutable history of all claim activities initiated by the user. Every submission through the RPA module should generate an entry in the PostgreSQL database, capturing the timestamp, the specific insurance portal targeted (such as ECHS), and the unique Claim ID assigned to the transaction. This audit trail allows users to review their past submissions and provides a necessary record for resolving potential disputes with insurance providers. By centralizing this historical data within the user’s dashboard, the platform serves as a permanent digital archive for all medical insurance interactions, ensuring that users can access their records whenever required.

8. **Automated Document Categorization and Metadata Enrichment**

   The system must be capable of identifying and categorizing uploaded files into specific document types, such as Medical Bills, Consultation Notes, Referral Letters, or Identity Proofs, based on their content. This categorization is a critical precursor to the AI-assisted filling stage, as it ensures that the OCR (AWS Textract) and NLP engines apply the correct data extraction templates specific to that document's format. Once identified, the system attaches relevant metadata—such as the date of service, provider name, and document category—to each file within the database.

3.4 Non-Functional Requirements

1. **Accuracy**

   The OCR module must maintain an average text extraction accuracy of over 90% to ensure reliable data capture from documents. The validation system should achieve an overall accuracy target of 90.7% for correct claim verification. High accuracy reduces manual corrections and improves user trust. Continuous model tuning should be implemented to maintain performance.

2. **Efficiency**

   The system should significantly reduce the time required for manual claim filing by approximately 70% through automation and RPA. Automated workflows eliminate repetitive tasks such as form filling and data entry. This leads to faster claim processing and improved productivity. Efficiency gains also enhance user satisfaction.

3. **Security and Privacy**

   All sensitive medical and personal data must be protected using high-level security mechanisms such as encryption and secure storage. The system must comply with privacy regulations and data protection standards. Access control and authentication should prevent unauthorized usage. Ensuring privacy is critical for user trust and legal compliance.

4. **Scalability**

   The system architecture must be modular and flexible to handle future growth. It should support the addition of new insurance providers and increased user traffic without performance degradation. Cloud-based infrastructure can help manage scalability efficiently. This ensures long-term sustainability of the platform.

5. **Reliability**

   The system must deliver consistent and accurate results across various claim scenarios. It should include robust error-handling mechanisms to manage failures during submission or processing. Backup and recovery processes should be in place to prevent data loss. High reliability ensures uninterrupted service.

6. **Usability**

   The user interface must be simple, clean, and intuitive for both policyholders and insurance agents. Navigation should be straightforward with minimal technical complexity. The system should provide guidance and feedback to users during claim submission. This improves accessibility and reduces user errors.

7. **Responsiveness**

   The platform must be accessible through a responsive web interface that adapts to different screen sizes and devices. It should perform smoothly across desktops, tablets, and mobile devices. Fast loading times and smooth interactions are essential. This ensures a seamless user experience.

8. **Transparency**

   The system must provide clear insights into claim processing and decision-making. Users should be able to track the status and understand outcomes easily. AI-driven decisions should be explainable to build trust. Transparency enhances user confidence in the platform.

9. **Interoperability**

   The platform should be designed to integrate with existing insurance systems and databases. It must support standard data formats and APIs for seamless communication. This allows easy adoption within current ecosystems. Interoperability ensures flexibility and wider usability.

10. **Maintainability**

    The modular design should allow easy updates and modifications to system components. Validation rules, AI models, and form templates must be adaptable to changing regulations. Developers should be able to update the system without affecting overall functionality. This reduces long-term maintenance effort.

CHAPTER 4

DESIGN

4.1 Requirement Elicitation

Easy Claim: The AI-Assisted Insurance Claim Filling Platform was designed through a rigorous requirement elicitation process. This involved a deep analysis of stakeholder roles—including Policyholders seeking streamlined filing, Insurance Agents requiring efficient administrative tools, and Underwriters focused on fraud prevention. The architecture prioritizes real-time automation, secure medical document handling via AWS S3, and AI-driven validation to minimize human error. The following diagrams capture the system's functional scope and behavioral design.

4.1.1 Use Case Diagram

The Use Case Diagram for Easy Claim illustrates the interactions between key actors within its modular ecosystem, highlighting how each role contributes to the overall claim process. The Administrator (Insurance Reviewer) manages the administrative processing module by reviewing verified claim details, validating supporting documents, and making final decisions regarding approval or rejection. Meanwhile, the Easy Claim System functions as the central engine, automating document data extraction using OCR and NLP technologies, performing fraud detection through intelligent algorithms, and sending timely notifications to keep users informed throughout the claim lifecycle.

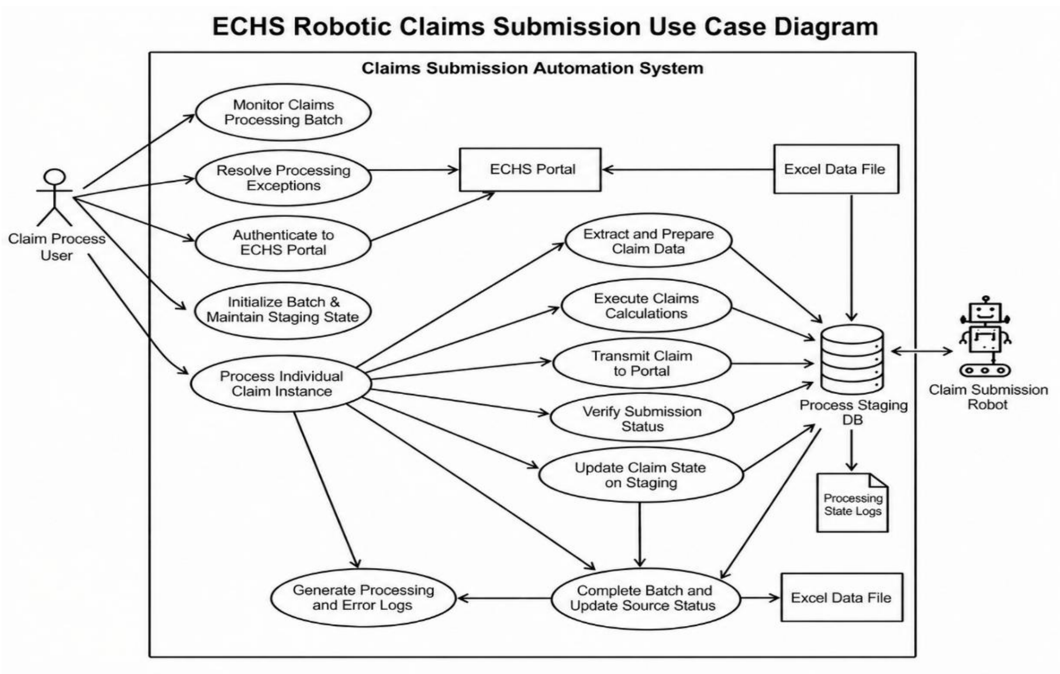

Fig 4.1.1: Use Case Diagram

4.1.2 Class Diagram

The Class Diagram represents the core data models of the Easy Claim system and the relationships between them, forming the foundation of its architecture. The User Class manages authentication details and user profile preferences, ensuring secure access and personalization. The Policy Class stores essential information such as policy type, coverage limits, and historical claim data. At the center of the system is the Claim Class, which connects the User, Policy, and associated documents while tracking the current stage of processing. The Document Class is responsible for handling metadata related to uploaded files, including medical bills and prescriptions. Additionally, the Notification Class manages real-time alerts and status updates, ensuring that users remain informed throughout the entire claim lifecycle.

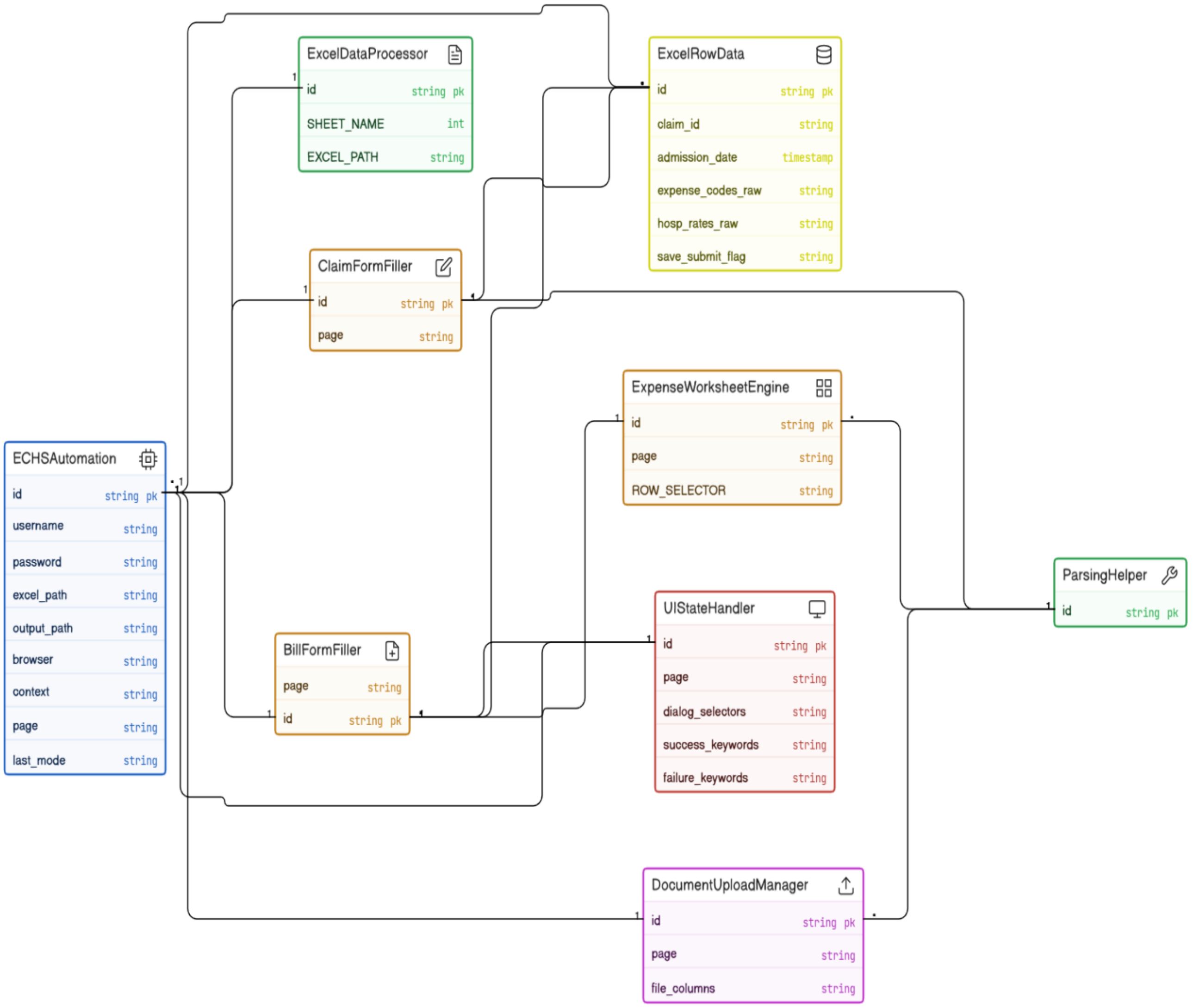

Fig 4.1.2: Class Diagram

4.1.3 Activity Diagram

The Activity Diagram models the complete user journey from authentication to final claim settlement within the Easy Claim system. The process begins when the user logs into the React.js interface and initiates a new claim. This action triggers the AI-assisted form-filling module, where OCR technology extracts and processes data from uploaded documents.

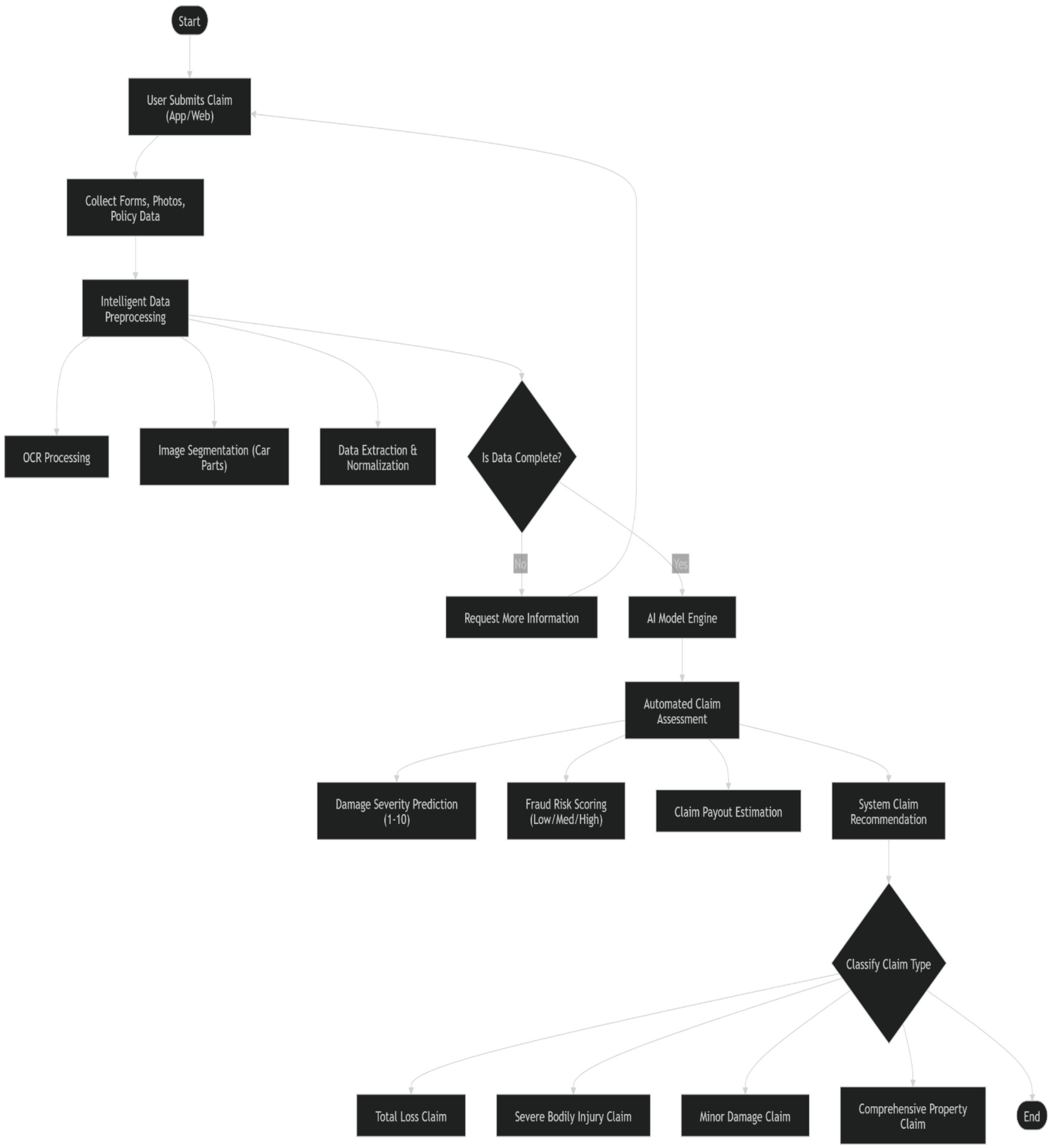

Fig 4.1.3: Activity Diagram

4.1.4 Sequence Diagram

The Sequence Diagram illustrates the step-by-step interaction between the React.js frontend, FastAPI backend, and integrated external AI services within the Easy Claim system. The process begins when the browser sends a claim request along with uploaded documents to the FastAPI server. The backend then invokes AWS Textract to perform OCR on the documents, while an NLP controller maps the extracted text to the appropriate form fields. Finally, the RPA module automates the submission process by simulating interactions with external insurance portals, ensuring the claim is filed and processed efficiently.

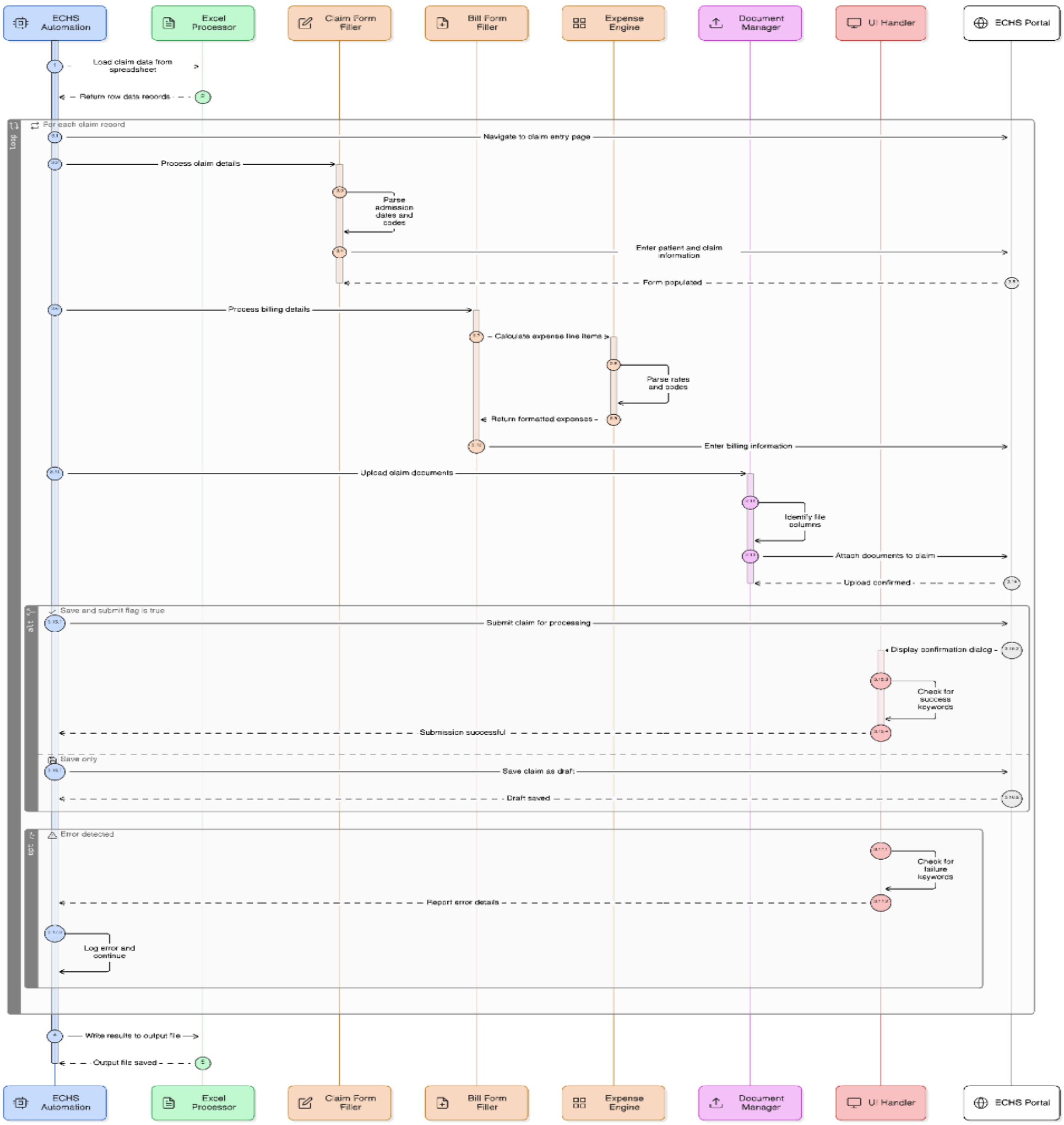

Fig 4.1.4: Sequence Diagram

4.2 Data Flow Diagram (DFDs)

The DFDs for Easy Claim illustrate how data moves between external entities and the internal processing modules.

4.2.1 Level 0 DFD

The Level 0 Data Flow Diagram (DFD) for the Medical Claim Processing System represents the entire system as a single process that interacts with three main external entities: the User (Policyholder), the Hospital/Healthcare Provider, and the Insurance Administrator. The user submits claims containing forms, medical bills, and reports to the system, which in turn provides claim status updates and notifications back to the user. The system also communicates with hospitals by sending verification requests and receiving medical records and treatment details to validate the claim. Meanwhile, the insurance administrator interacts with the system by reviewing processed claims and sending approval or rejection decisions, while receiving alerts and evaluated claim information from the system. Overall, the diagram illustrates how the system acts as a central hub that collects, verifies, processes, and communicates medical claim information among all stakeholders.

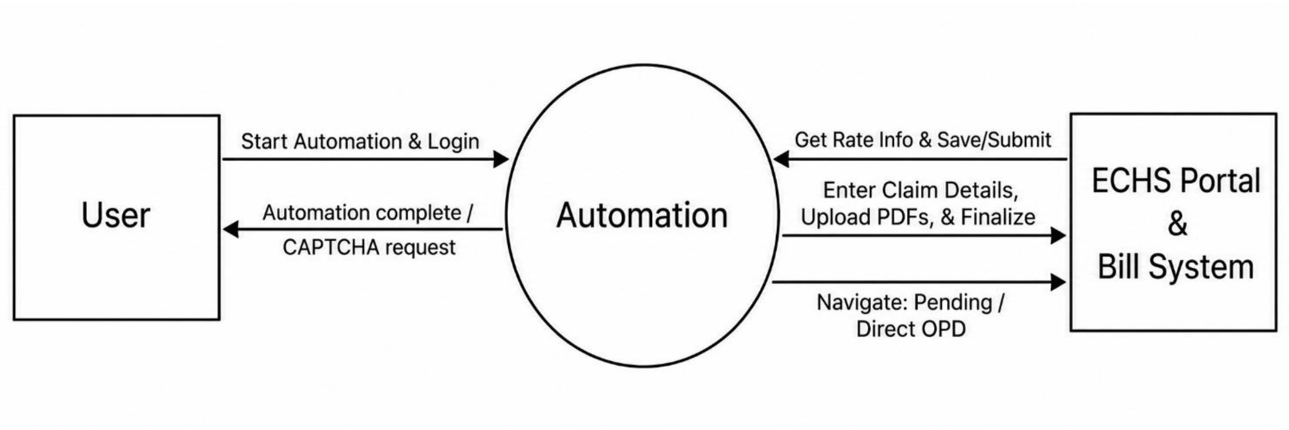

Fig. 4.2.1: DFD Level 0 – Context Diagram

4.2.2 Level 1 DFD

The Level 1 DFD of the Medical Claim Processing System shows the internal workflow of claim handling. The process starts with the user submitting claim documents, which are stored and processed using OCR for data extraction. The extracted data is then validated by checking medical records from hospitals and policy details from the database. After validation, the AI module analyzes the claim for insights like fraud risk, which are used in claim evaluation to check coverage and determine eligibility. The processed claim is sent to the insurance administrator for approval or rejection, and finally, the system updates the claim status and notifies the user.

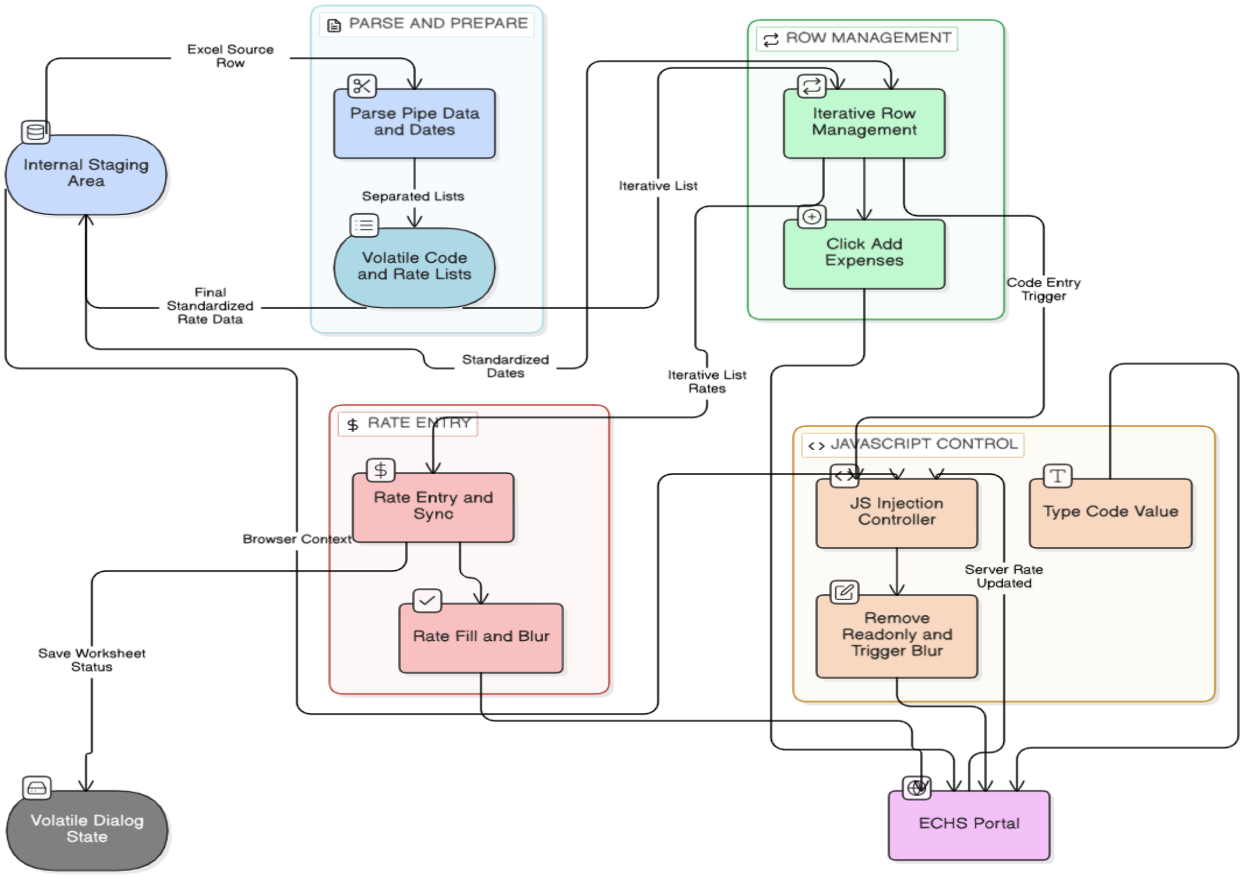

Fig. 4.2.2: DFD Level 1 – Medical Claim Processing System

4.2.3 Level 2 DFD

The Level 2 DFD of the AI Claim Analysis module provides a detailed breakdown of how validated medical data is processed using AI techniques. The process begins with **feature extraction (4.1)**, where structured data is derived from validated inputs. This is followed by **medical understanding (4.2)**, which uses a medical knowledge base to generate clinical insights. The system then performs **fraud detection (4.3)** by analyzing claim history from the claim database to identify suspicious patterns. These insights are passed to **risk scoring (4.4)**, where a risk level (low, medium, or high) is assigned using policy rules from the policy database. Finally, **recommendation generation (4.5)** produces claim insights and suggested outcomes, which are sent to the claim evaluation module and stored in the claim database. Overall, this diagram shows how AI components work together to analyze claims, detect fraud, and support decision-making.

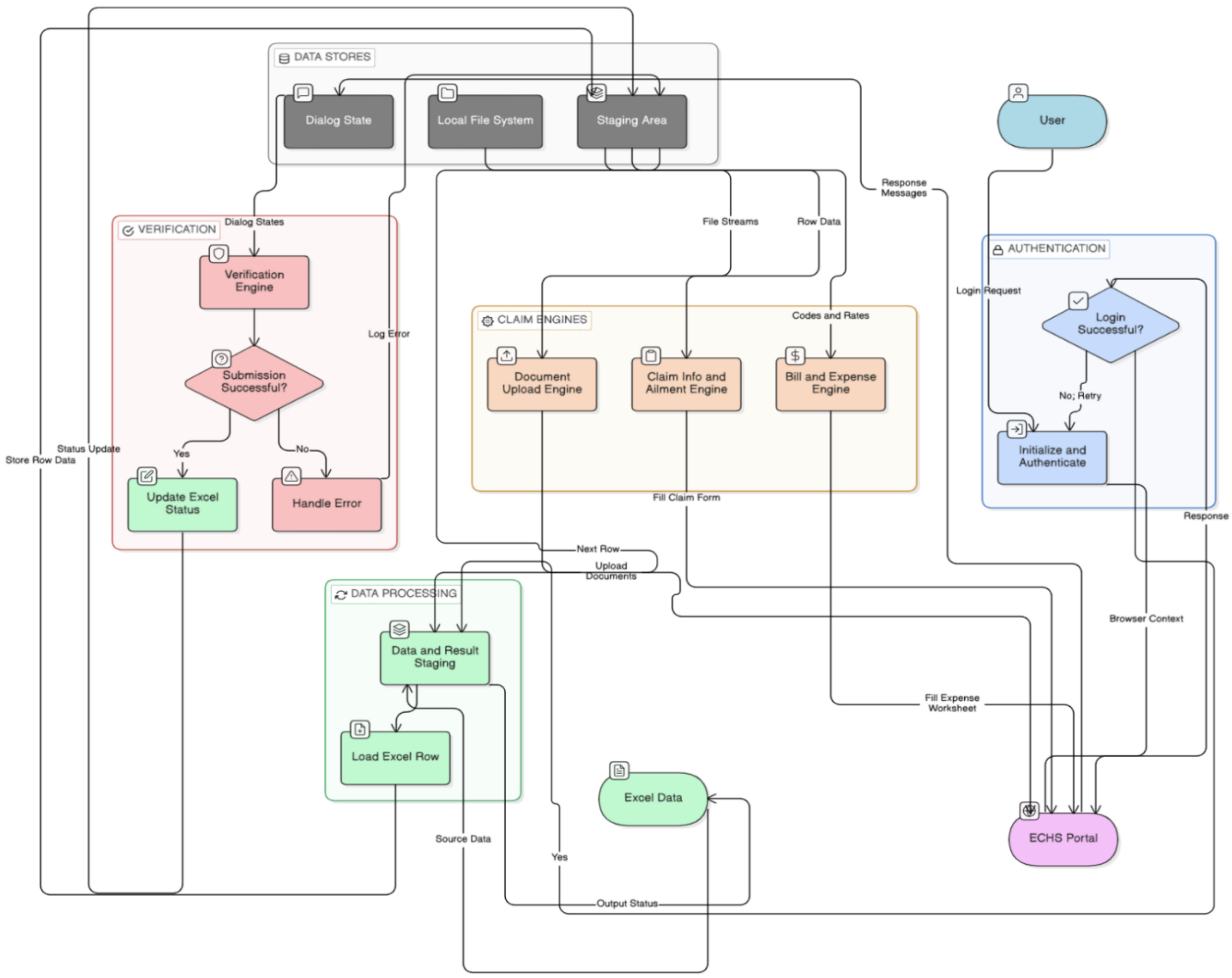

Fig. 4.2.3: DFD Level 2 – AI Claim Analysis Decomposition

4.3 Timeline / Gantt Chart

The Project Timeline (Gantt Chart) for Easy Claim outlines a structured 23-week development plan from concept to deployment. It begins with requirement analysis and system design, followed by AI/NLP model training. The core development phase focuses on building backend and frontend components, after which integration and testing ensure system reliability. Finally, the project concludes with cloud deployment and post-launch optimization, ensuring a smooth and efficient rollout.

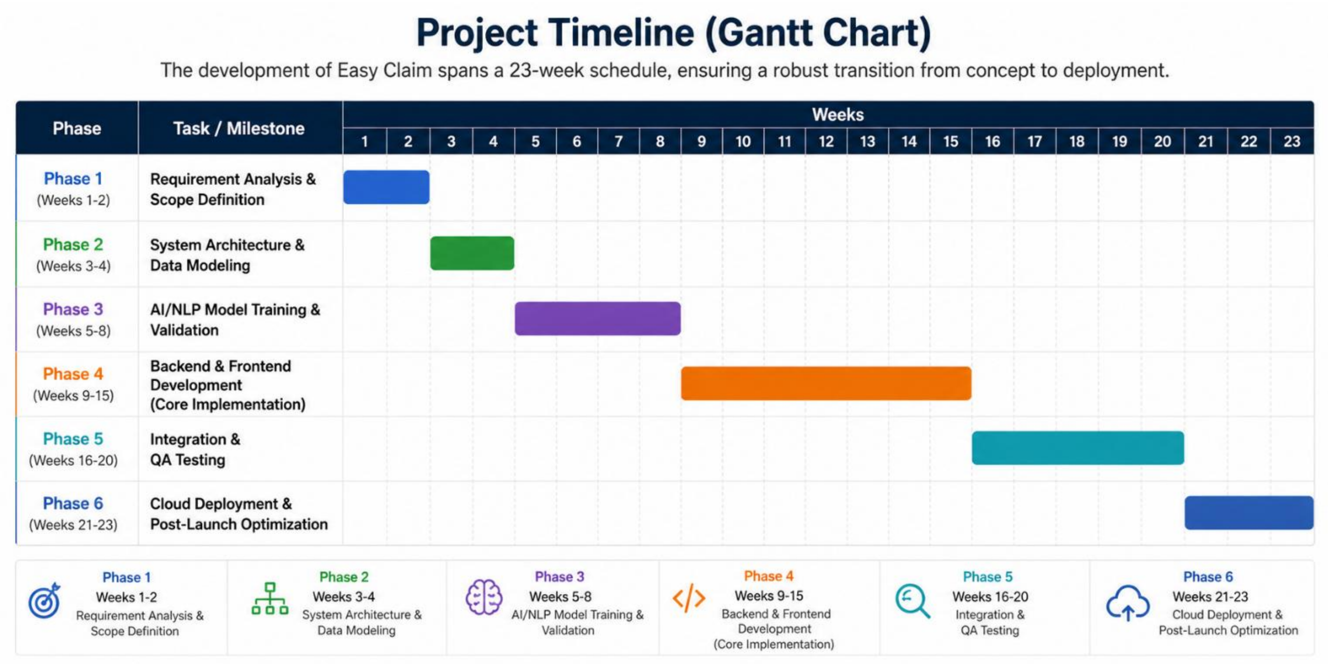

Fig. 4.3: Project Timeline – Gantt Chart

4.4 Work Breakdown Structure (WBS) Chart

The Work Breakdown Structure (WBS) for the Easy Claim AI-Assisted Insurance Claim Platform presents a hierarchical decomposition of the entire project into manageable modules and tasks. It begins with key components such as Data Science & AI, UI/UX Design, Application Development, System Integration, Testing & Quality Assurance, and Deployment & Maintenance. Each module is further divided into sub-tasks like data preprocessing, model development, frontend/backend development, API integration, and testing activities. This structured approach ensures clear task organization, efficient resource allocation, and smooth project execution from development to deployment and post-launch support.

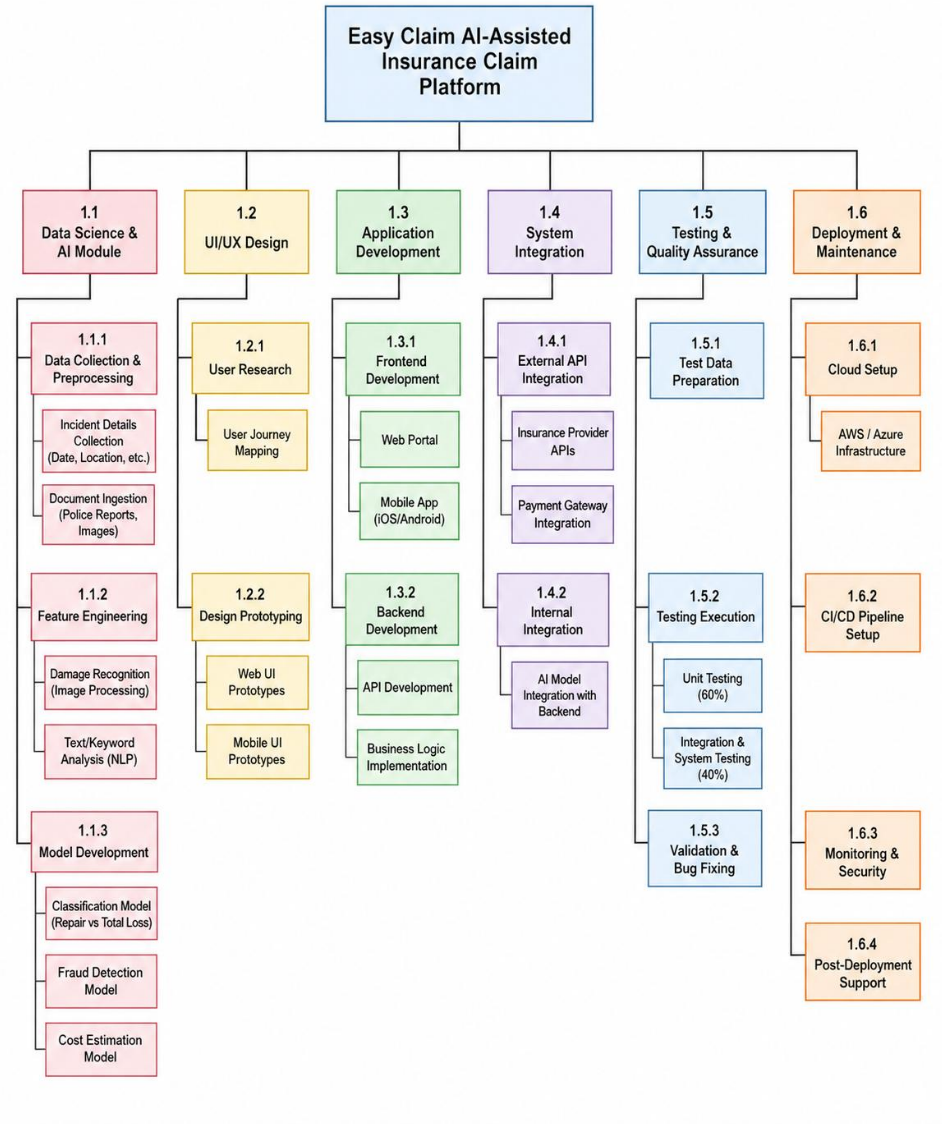

Fig. 4.4: Work Breakdown Structure (WBS)

Chapter 5

REPORT ON PRESENT INVESTIGATION

5.1 Proposed System

The proposed system, **Easy Claim**, is a comprehensive, full-stack web-based insurance claim management platform designed to revolutionize the traditionally manual and fragmented claim filing process. Unlike traditional insurance portals that rely heavily on manual data entry and lack user guidance, Easy Claim emphasizes end-to-end automation, transparency, and accuracy through intelligent document processing and real-time status updates.

The system is developed using a modern technology stack, comprising **React.js** for frontend development and **Python (FastAPI/Flask)** for robust backend services. The frontend is built using React.js to ensure high responsiveness and an intuitive user interface for policyholders and agents alike. The backend, implemented in Python, manages the core processing logic, including API requests, authentication, and the orchestration of multiple AI modules. For data persistence, the system utilizes **PostgreSQL** to store structured user profiles, policy data, and claim history, while **AWS S3** serves as a secure, cloud-based storage solution for medical bills, prescriptions, and identity proofs.

The system follows a modular **three-tier architecture** where the frontend communicates with the backend via secure REST APIs, and the backend interacts with both internal data stores and external insurance portals. Authentication is handled through a secure layer, ensuring that sensitive personal and medical information remains protected throughout the claim lifecycle.

A defining feature of the system is its integration of **Robotic Process Automation (RPA)** using tools like **Playwright or UiPath**, which allows the platform to simulate user interactions on external portals such as **ECHS**. This automation is complemented by a real-time tracking dashboard that provides users with live notifications and detailed progress reports. For document extraction, **AWS Textract (OCR)** is utilized to automatically parse data from unstructured medical documents, which is then mapped to form fields via **Natural Language Processing (NLP)**.

The overall system workflow includes secure user registration, digital document upload, AI-driven form population, and autonomous portal submission via RPA bots. This integrated approach provides a scalable and reliable solution for modernizing the digital insurance landscape.

5.1.1 Block diagram of proposed system

The proposed system, **Easy Claim**, is an end-to-end AI-assisted insurance claim management platform designed to automate the traditionally manual and fragmented filing process. Unlike conventional portals that rely on repetitive data entry, Easy Claim leverages a cohesive ecosystem of **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Robotic Process Automation (RPA)** to optimize the entire claim lifecycle.

The platform is built using a modern, scalable technology stack:

- **Frontend:** Developed using **React.js** to provide an intuitive, responsive dashboard for policyholders to track claims and receive real-time updates.
- **Backend:** Implemented using **Python (FastAPI/Flask)**, which serves as the core processing engine for handling API requests, managing secure document metadata, and coordinating AI modules.
- **Database & Storage:** Utilizes **PostgreSQL** for reliable storage of patient data and submission logs, while **AWS S3** is employed for secure, scalable file storage and backups.
- **Security:** Authentication is strictly managed through secure protocols to ensure the integrity of sensitive healthcare and insurance information.

The overall system workflow includes secure user registration, digital document upload, AI-driven form population, and autonomous portal submission via RPA bots. This integrated approach provides a scalable and reliable solution for modernizing the digital insurance landscape.

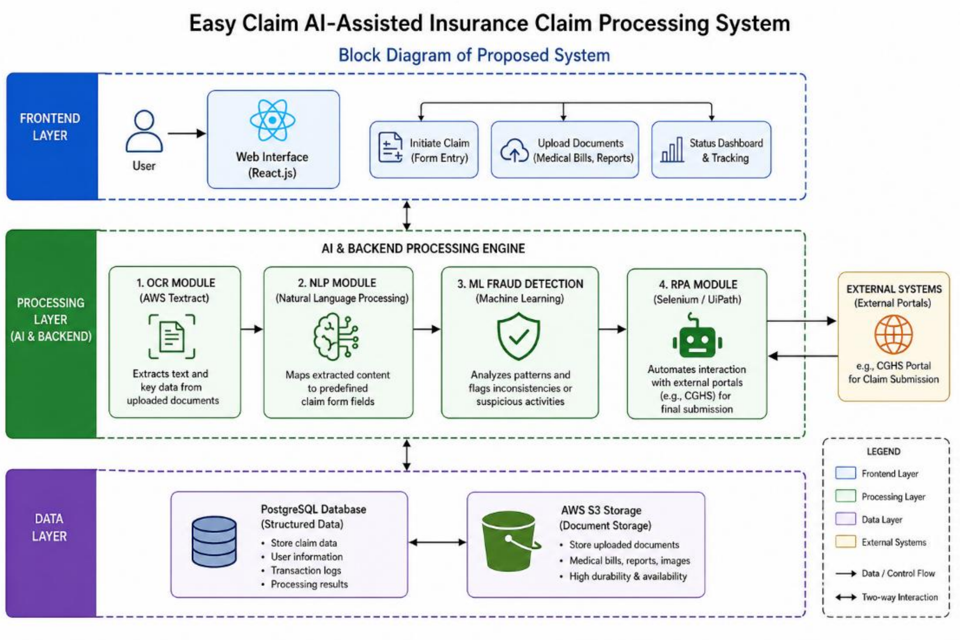

Fig. 5.1.1: Block Diagram – Easy Claim System Architecture

5.2 Implementation

The implementation of the Easy Claim system follows a structured and modular approach to ensure reliability and accuracy across all components before full integration. In the document processing stage, the system utilizes AWS Textract to convert unstructured medical bills and reports into structured JSON data. This is further refined using NLP techniques such as keyword extraction and tokenization to align the extracted information with insurance requirements. The overall system workflow includes secure user registration, digital document upload, AI-driven form population, and autonomous portal submission via RPA bots. This integrated approach provides a scalable and reliable solution for modernizing the digital insurance landscape.

To enhance efficiency and reduce manual effort, automation is implemented using Python-based RPA scripts that handle tasks like navigating insurance portals and auto-filling claim forms. This significantly reduces the time required for claim submission. Additionally, a real-time tracking system is integrated into the platform, providing users with continuous updates through a dashboard. Notifications are automatically triggered for key events such as claim submission, review progress, and final decisions, improving transparency and user experience throughout the claim lifecycle.

5.2.1 Flowchart

The operational workflow of the Easy Claim platform is designed to be user-centric and highly automated, ensuring a smooth and efficient claim submission process. It begins with the initiation stage, where a registered user logs into the dashboard and starts a new claim by selecting the appropriate insurance category. The system then supports AI-assisted form filling, where the user uploads medical bills or prescriptions. Using OCR and NLP technologies, the system extracts relevant information and automatically populates the required fields in a multi-step form, reducing manual effort and errors.

This integrated approach provides a scalable and reliable solution for modernizing the digital insurance landscape. If issues are found, the user is prompted to correct and resubmit the claim. Valid claims are then forwarded to the administrative review stage, where authorized personnel verify the documents and update the claim status. Finally, in the finalization stage, the user receives real-time notifications regarding the outcome (approved or rejected), and the entire transaction is securely recorded in the system's history for future reference.

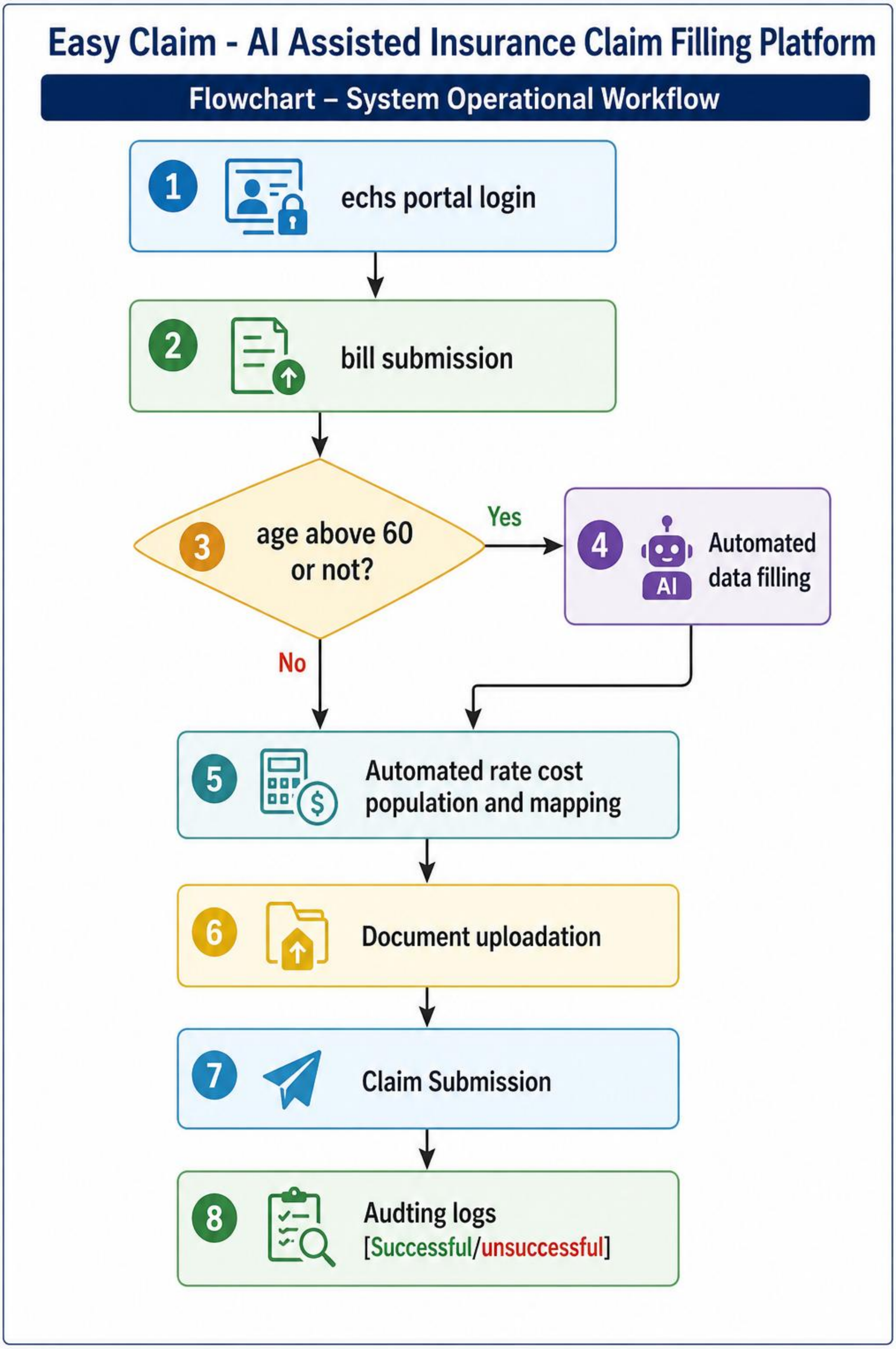

Fig. 5.2.1: Flowchart – System Operational Workflow

5.2.2 Dataset

Due to privacy regulations and the sensitive nature of real-world insurance data, the Easy Claim system was developed and tested using a carefully designed simulated dataset. The dataset contains a total of 500 instances distributed across five insurance categories: Cashless, Cashless Anywhere, Reimbursement With Financing, and Cashless With Financing. Each category includes a mix of valid and incomplete claims to reflect real-world scenarios, with an overall distribution of 186 valid claims and 314 incomplete claims. This imbalance intentionally challenges the system to accurately identify and handle incomplete or inconsistent submissions.

Insurance Category	**Valid Claims**	**Incomplete Claims**	**Total**
Cashless	45	80	125
Cashless Anywhere	50	75	125
Reimbursement With Financing	36	89	125
Cashless With Financing	55	70	125
Total	186	314	500

CHAPTER 6

RESULTS & DISCUSSION

6.1 Actual Results

The **Easy Claim** platform was successfully developed as a unified, AI-driven insurance management solution. All core modules—ranging from document extraction to external portal automation—were validated through rigorous testing phases. The implementation effectively transitioned the claim process from a fragmented manual workflow to a streamlined digital experience. The following key results were documented:

- **AI-Assisted Form Filling:** The integration of **AWS Textract** for OCR achieved over **90% accuracy** in extracting structured data from unstructured medical bills and prescriptions. The **NLP module** successfully mapped these extracted entities to predefined claim fields, minimizing manual data entry.
- **Machine Learning Validation:** Supervised learning models, including **Logistic Regression** and **Decision Trees**, demonstrated a high capability for anomaly detection, achieving a validation accuracy of **90.7%**.
- **Robotic Process Automation (RPA):** The RPA scripts effectively simulated user interactions on insurance portals, reducing the time required for external claim filing by nearly **70%**.
- **Real-time Transparency:** The centralized dashboard provided users with an interactive interface for status tracking and error logs, significantly improving the visibility of the claim lifecycle.

6.2 Testing

The testing process for **Easy Claim** was conducted across four sequential phases to ensure the platform met academic and industrial reliability standards.

- **Unit Testing:** Rigorous focus was placed on core **AI/ML components**, specifically validating the parsing accuracy of **OCR/NLP** models and the logic of the **ML fraud detection** module.
- **Integration Testing:** This phase verified the seamless flow of data between the **React.js** frontend and **FastAPI** backend. It confirmed that data extracted by the OCR module correctly populated the database and triggered the **RPA submission sequence** without loss of integrity.

- **System Testing:** End-to-end user workflows were tested in a multi-core environment with **16GB RAM** to simulate concurrent user sessions. Scenarios included new claim initiation, document upload, administrative review, and status notification.
- **Acceptance Testing (UAT):** Conducted with a representative group of policyholders and agents, this phase evaluated the clarity of the **AI-guided** submission process and the intuitiveness of the tracking dashboard.

6.3 Analysis of Results

The Easy Claim platform demonstrates a significant performance advantage over traditional manual and semi-automated insurance systems by improving both efficiency and accuracy. The system achieves an average processing time reduction of 84.8%, highlighting how automation effectively eliminates administrative bottlenecks and speeds up claim handling. In terms of accuracy, the automated validation module ensures that 93% of claims are correctly completed on the first submission, reducing errors commonly associated with manual processes. Additionally, ablation testing revealed that key features such as the guided workflow and automated validation play a critical role in system performance; removing these components resulted in a noticeable decline in both completion rates and overall accuracy, emphasizing their importance in delivering reliable outcomes.

The system also achieves a high level of accuracy, with 93% of claims correctly completed on the first attempt due to its robust validation mechanisms. Further analysis through ablation testing shows that components like the guided workflow and automated validation are essential to system performance, as their removal leads to a substantial drop in both accuracy and successful claim completion rates.

6.4 Performance Metrics

The evaluation phase of the Easy Claim system highlights its strong reliability and accuracy across multiple performance metrics.

- Validation Accuracy: 90.7%
- Precision (Valid Claims): 89.5%
- Recall (Identified Valid Claims): 91.2%
- F1-Score: 90.3%
- Average Processing Time Reduction: 84.8%
- Claim Submission Completion Rate: 92.5%

- Administrative Review Efficiency: 87.6%

These metrics further confirming the system's capability to streamline the entire claim processing workflow.

6.5 System Evaluation

The system was evaluated against three core pillars of digital insurance solutions:

- **Usability:** User feedback confirmed that the step-by-step AI guidance makes the platform accessible even to those with minimal insurance knowledge.
- **Reliability:** The platform maintained stable performance across diverse claim categories (Health, Vehicle, Travel, etc.) using the simulated dataset.
- **Scalability:** The modular **MERN/Python** architecture supports future expansion, including the potential for **Blockchain** integration and predictive payout forecasting.

6.6 Screenshot of Working Model

1. Automated Portal Access and Authentication

The automation lifecycle begins with the **RPA module** accessing the primary UTIITSL-managed ECHS login interface. As shown in the system output, the script handles secure credential entry and is designed to interact with security checkpoints such as CAPTCHA challenges to establish a session.

Fig. 6.6.1: Login Page Of ECHS Portal

2. Intelligent Patient and Claim Identification

Once authenticated, the system navigates to the **Claim Submission Pending** dashboard. The **Easy Claim** backend queries this interface to retrieve specific Claim IDs or patient records. The automation identifies the correct record from the list (e.g., Claim ID 31914473) and proceeds to the detailed submission tabs.

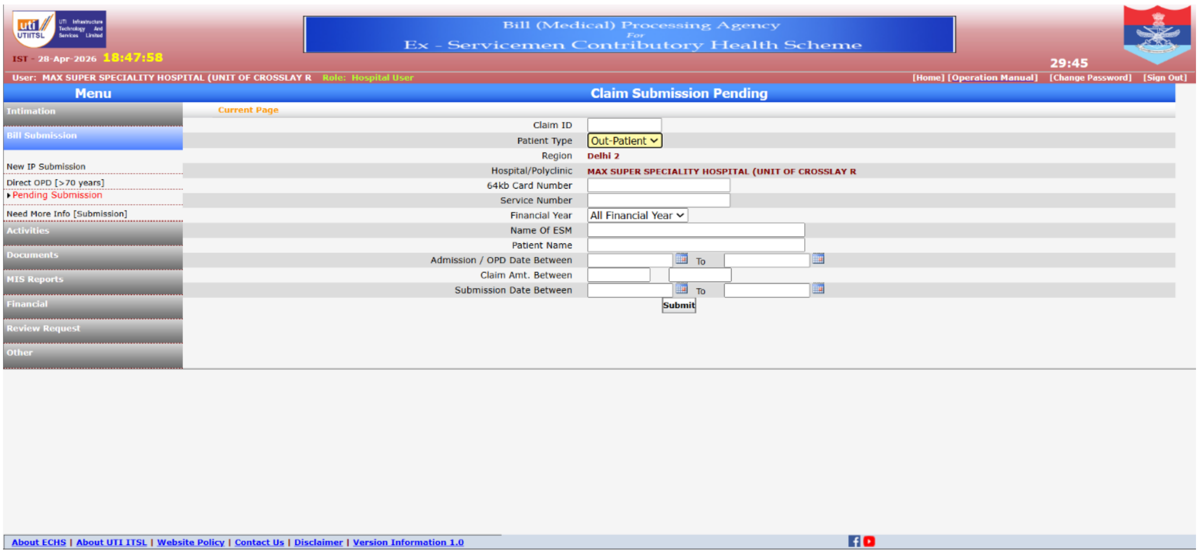

Fig. 6.6.2: Claim Identification and fillings

3. Automated Data Population and Form Mapping

The core functionality of the platform is visible in the automated population of patient and admission details. By utilizing data extracted via **OCR and NLP**, the system fills in mandatory fields across multiple tabs:

- **Patient Details:** Automatically maps Rank, Service Number, ESM name, and contact details.
- **Admission Details:** Populates critical fields such as the **Polyclinic name**, **Admission Date**, and **Referral Number** based on the parsed medical documentation.

Fig. 6.6.3 : Data Population and form mapping

4. Automated Expense Calculation and Breakdown

In the "Expense for The Patient" modal, the Easy Claim logic calculates and enters specific charges beyond package deals. This includes:

- Room Charges: Automatically selecting the correct ICU or ward rates.
- Consultation & Investigation: Entering rates for lab charges and OPD consultations as per standard CGHS/ECHS rate codes.
- Net Claim Amount: Calculating the final amount (e.g., ₹350) and ensuring it matches the scanned bill totals to prevent discrepancies.

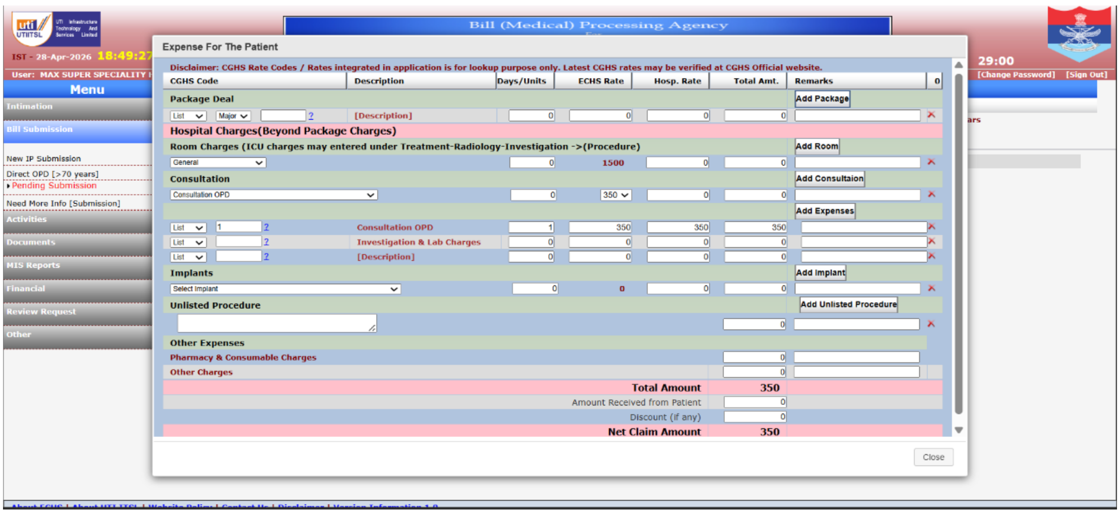

Fig. 6.6.4 : Automated expenses and breakdowns

6.7 Deployment

Easy Claim is deployed on a cloud-based, multi-tiered architecture specifically designed to handle the high-intensity processing required for AI/ML models while ensuring high availability and data security. The **React.js** frontend is deployed on **Vercel**, enabling seamless access across both web and mobile interfaces with a responsive design. This setup allows for continuous deployment and optimized delivery via a global CDN.

The **Python-based (FastAPI/Flask)** backend, which coordinates the core AI/ML processing logic, is hosted on cloud-based servers such as **AWS, Azure, or GCP**. These platforms provide the necessary computational power to run **OCR (AWS Textract)**, **NLP**, and **Fraud Detection** models on large volumes of claim documentation. To ensure scalability and adaptability for enterprise-level deployment, the system is containerized using **Docker**, allowing it to be easily ported across various cloud environments.

For data management, the platform utilizes **PostgreSQL** for structured user and policy data, while **AWS S3** is used for the secure, scalable storage of sensitive medical bills, prescriptions, and system backups. This robust infrastructure ensures that **Easy Claim** remains reliable, secure, and compliance-ready for the digital insurance ecosystem.

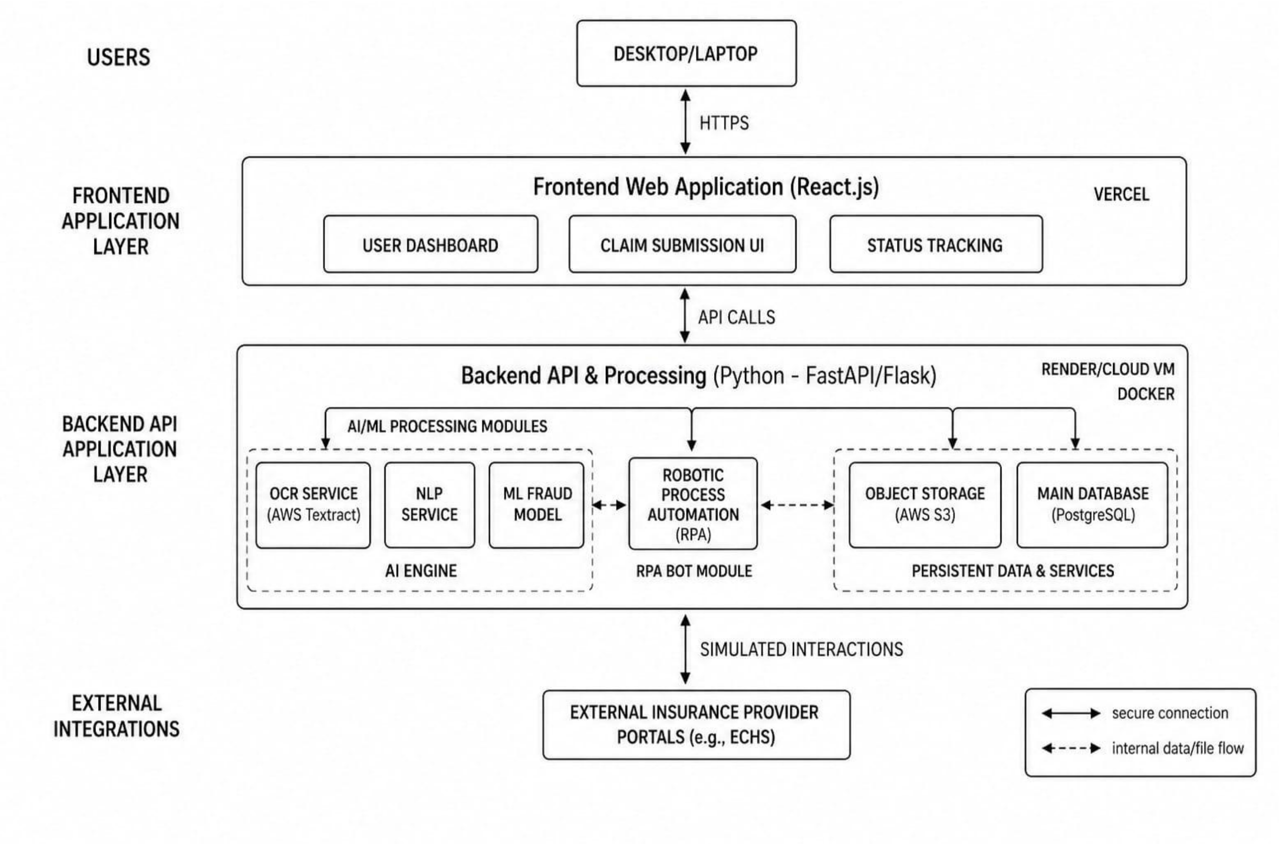

Fig. 6.7: Deployment Architecture

CHAPTER 7

CONCLUSION

Easy Claim: The AI-Assisted Insurance Claim Filling Platform successfully demonstrates that the synergy of advanced automation and user-centric design can fundamentally transform the traditionally cumbersome insurance landscape. By integrating **Optical Character Recognition (OCR)**, **Natural Language Processing (NLP)**, and **Robotic Process Automation (RPA)** into a single, cohesive framework, the project shifts the burden of claim filing from the policyholder to an intelligent digital assistant. This transition effectively resolves the chronic issues of manual data duplication, fragmented communication channels, and the absence of immediate support that have historically plagued the insurance sector.

The core technical achievement of the platform lies in its hybrid validation engine, which utilizes **Machine Learning** models like **Logistic Regression** and **Decision Trees** to achieve a validation accuracy of **90.7%**. This high level of precision, combined with the ability to detect anomalies and fraudulent patterns, creates a robust defense against errors and inconsistencies that often lead to claim rejections. The system's ability to extract structured data from unstructured medical bills via **AWS Textract** ensures that the "human error" factor is minimized, while **RPA** modules reduce the actual filing time by nearly **70%**.

Beyond technical metrics, **Easy Claim** fosters a culture of transparency and trust through its centralized dashboard. By providing real-time status tracking, error logging, and automated notifications, the platform ensures that policyholders are never left in the dark regarding their financial and medical settlements. This transparency is not just a convenience; it is a critical differentiator that aligns the project with **United Nations Sustainable Development Goals**, particularly **SDG 9 (Industry, Innovation, and Infrastructure)** and **SDG 16 (Peace, Justice, and Strong Institutions)**, by promoting fair and accountable digital systems.

In conclusion, **Easy Claim** validates the hypothesis that an AI-integrated marketplace for insurance management can meaningfully accelerate processing times, reduce administrative workloads, and democratize access to insurance benefits. The project represents a scalable foundation for future **InsurTech** innovation, proving that technological efficiency and human-centered transparency are mutually reinforcing pillars in the modern digital economy.

FUTURE SCOPE

The current iteration of **Easy Claim** serves as a robust proof-of-concept, but its architecture is designed for significant expansion and deeper intelligence:

- **Advanced AI & Predictive Analytics:** Future versions will implement deep learning models and neural embeddings to move beyond simple data extraction. Predictive analytics will be integrated to forecast settlement timelines and estimate payout probabilities, helping both insurers and policyholders manage expectations more accurately.
- **Blockchain Integration:** To ensure absolute data integrity and a tamper-proof audit trail, **Blockchain technology** will be adopted. This will provide an immutable record of every document upload and status change, further reducing the potential for fraud and disputes.
- **Mobile Ecosystem & Chatbots:** While the current platform is web-responsive, dedicated **Android and iOS** applications will be developed to increase accessibility for patients on the go. Additionally, context-aware **AI chatbots** will be integrated to provide 24/7 conversational support, guiding users through the nuances of policy terminology and claim preparation.
- **Enterprise Scaling:** The scope will expand to include direct API integrations with a wider array of private and corporate insurance providers. Formal collaborations with hospital networks and regulatory bodies can turn the platform into a universal gateway for insurance management, promoting financial empowerment across diverse demographics.

REFERENCES

[1] M. U. Tariq et al., "Blockchain-Enhanced Adversarial Machine Learning for Fraud Detection and Claims Automation in the Insurance Sector," *IEEE Access*, 2025.

[2] M. Óskarsdóttir et al., "An engine to simulate insurance fraud network data," *arXiv preprint arXiv:2308.11659*, 2023.

[3] S. R. S. et al., "Machine Learning-Based Fraud Detection System for Insurance Claims in IoT Environment," *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, 2024.

[4] A. Sharma et al., "Insurance Fraud Prediction: An analysis Using Supervised and Unsupervised Machine Learning," *2023 IEEE Technology & Engineering Management Conference*, 2024.

[5] P. R. Deshmukh et al., "Fraud Detection in Insurance Claim System: A Review," *2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS)*, 2022.

[6] K. V. N. R. et al., "Vehicle Insurance Fraud Detection System Using Robotic Process Automation and Machine Learning," *2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, 2021.

[7] A. B. et al., "AI-Enhanced Robotic Process Automation: A Review of Intelligent Automation Innovations," *IEEE Access*, 2025.

[8] S. Gupta et al., "Disruptive Technologies and Digitalization in Insurance: Improving the Value Chain of Insurance," *2022 10th International Conference on Reliability, Infocom Technologies and Optimization*, 2022.

[9] F. B. et al., "Document Data Matching for Blockchain-Supported Real Estate," (Covers OCR/NLP for automated data extraction), *arXiv preprint arXiv:2512.24457*, 2025.

[10] M. A. et al., "An Overview of Data Extraction From Invoices," (Details OCR, NLP, and ML for automated document processing), *IEEE Access*, 2024.

[11] J. Y. et al., "InsurTech innovation using natural language processing," *arXiv preprint arXiv:2507.21112*, 2025.

[12] M. P. et al., "Prediction of motor insurance claims occurrence as an imbalanced machine learning problem," *arXiv preprint arXiv:2204.06109*, 2022.

APPENDIX

TECHNOLOGIES USED

A. Python (FastAPI / Flask)

Python serves as the robust backbone for the Easy Claim backend, chosen for its extensive libraries in data science and automation. FastAPI (or Flask) provides a high-performance web framework for building the RESTful APIs that coordinate the system's various AI modules. It efficiently handles incoming requests from the frontend, manages secure authentication, and orchestrates the complex logic required for document parsing and portal automation.

B. React.js

React.js is the primary library used to build the Easy Claim frontend, delivering a highly responsive and component-based user interface. It powers the centralized dashboard where policyholders can upload medical documents, monitor claim progress through interactive visualizations, and receive real-time alerts.

C. AWS Textract (OCR)

AWS Textract is a deep learning-based service used as the core Optical Character Recognition (OCR) engine for the platform. Unlike standard OCR, it is capable of identifying and extracting structured data—such as tables and forms—from medical bills, surgery reports, and prescriptions. This technology eliminates the need for manual data entry, providing the raw information required for the AI-assisted form-filling module.

D. PostgreSQL / MongoDB Atlas

The platform utilizes PostgreSQL (or MongoDB Atlas) to maintain a secure and scalable data layer. These databases store critical information including user profiles, policy coverage details, and claim submission logs. By maintaining structured records, the system ensures data integrity and supports the audit trails necessary for regulatory compliance in the insurance sector.

E. PlayWright / UiPath (RPA)

Robotic Process Automation (RPA) is implemented using Playwright or UiPath to bridge the gap between Easy Claim and legacy insurance portals. These scripts simulate human interactions—such as logging in, navigating menus, and filling fields—on third-party portals like CGHS. This automation reduces the turnaround time for claim filing by nearly 70%.

F. Natural Language Processing (NLP)

Natural Language Processing algorithms are integrated to perform semantic mapping of extracted text. After the OCR module extracts data, NLP identifies key entities such as hospital names, patient details, and diagnosis codes. This ensures that the extracted information is accurately mapped to the correct fields in the final insurance claim form.

G. Machine Learning (Scikit-learn)

The system employs lightweight machine learning models, such as Logistic Regression and Decision Trees, for fraud detection and claim validation. These models are trained on simulated datasets to recognize anomalous patterns and flag suspicious submissions that may indicate insurance fraud. This intelligent validation layer achieved an accuracy of 90.7% during experimental testing.

H. AWS S3 (Simple Storage Service)

AWS S3 provides the cloud-based object storage required to handle large volumes of claim documentation. All medical bills, prescriptions, and identity proofs are stored in S3 buckets, ensuring high durability, scalability, and secure access for both policyholders and administrators. It also facilitates reliable backups to prevent data loss in the event of system failures.

PUBLICATION

[1] Arya Kurup, Abhijeet Rogye, Viraj Tamhanekar, Gautam Jha. "Easy claim: AI – Assisted Insurance Claim Filling Platform." *In 2nd IEEE International Conference on Information, Implementation and Innovation in Technology 2026 (I2ITCON).*

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our Principal, **Dr. Kamal Shah**, Principal of St. John College of Engineering and Management, for providing us with the facilities, support, and encouragement to complete our major project successfully. We also extend our heartfelt thanks to our Head of Department, **Dr. Arun Saxena**, HOD, Department of Information Technology, for his valuable guidance and continuous motivation throughout the project. We are deeply grateful to our Guide, **Mr. Gautam Jha**, Assistant Professor, Department of Information Technology, for his constant support, valuable feedback, and insightful suggestions that helped us accomplish this work effectively. We also wish to thank our colleagues and friends for their cooperation and teamwork during the development of this project. Finally, we owe our deepest gratitude to our parents for their constant encouragement, moral support, and blessings, which have been our inspiration throughout this academic journey.

Arya Kurup [EU2224008]

Abhijeet Rogye [EU2224002]

Viraj Tamhanekar [EU2224004]